

Digital Payments and Consumer Scam Risk

Main regression results, Stages 1 to 3 · Global Findex 2025 microdata · 27 September 2026

Results at a glance

	Outcome	N	β_W	95% CI	p	Reading
● Stage 1	Reported scam contact	16,251	+5.36	[+2.88, +7.83]	< 0.001	Significant, positive
● Stage 2	Sending money, conditional on contact	4,768	+1.71	[−1.03, +4.44]	0.218	Not significant
● Stage 3	Contact and sending money, joint event	16,249	+1.16	[+0.23, +2.10]	0.016	Significant, positive

β_W compares weekly payers with less-than-monthly payers in the same economy, in percentage points, from specification (2).

Wording for the paper, from the interpretation rule

higher reported contact with no detectable difference in conditional sending

Rule: Stage 1 is significant and Stage 2 is not, so the word “tradeoff” does not apply. Stage 3 is reported as the overall association.

Sample flow

Step	N	Economies	Weighted share of previous step
All respondents	144,090	140	
Economies where the scam module was fielded	76,898	75	53.4%
Owns a mobile phone (<code>con1</code> = 1)	64,663	75	81.4%
Paid in store by phone or card (<code>fin25e2</code> = 1)	16,655	75	24.8%
Base sample: valid <code>fin25e3</code> and <code>con21</code>	16,472	75	99.0%
Common sample: all controls present	16,251	75	98.5%

N is the unweighted number of respondents. All three stages start from the common sample.

Estimation

$$Y_{ic} = \beta_W \text{Weekly}_{ic} + \beta_M \text{Monthly}_{ic} + X'_{ic}\gamma + \alpha_c + \varepsilon_{ic}$$

- Weighted linear probability model, weights `wgt`.
- Country fixed effects α_c ; standard errors clustered by economy.
- Confidence intervals and p-values use the t distribution with 74 degrees of freedom.
- Reference group: less than monthly (`fin25e3 = 3`).
- The same equation, the same three specifications and the same sample within each stage.

Specification	Right-hand side
(1) Country FE only	Payment frequency and country fixed effects
(2) Main: demographics	(1) plus age, age squared / 100, female, secondary education, tertiary education, income quintile 2, income quintile 3, income quintile 4, income quintile 5, in the labor force, rural
(3) Plus digital engagement	(2) plus uses the internet, owns a smartphone, has an account

Notes for reading the results

- **Associations only.** Nothing here is a causal estimate.
- **Raw rates and regression estimates can point in different directions.** Economies with a high share of weekly payers have lower contact and sending rates overall. The raw rates in Tables 1.2, 2.2 and 3.2 mix this between-economy difference with the within-economy difference. The regressions use only the within-economy difference.
- **Predicted probabilities are point estimates.** Confidence intervals are reported for the differences only.
- **Stage 2 keeps all 75 economies.** Seven economies have no respondent who sent money. A linear probability model keeps them in the sample.
- **Lesotho.** Income is missing for the whole country. A constant is filled in and absorbed by the country fixed effect, so Lesotho stays in the sample.
- **Standard errors.** The small-sample correction counts the country dummies as parameters, as in Stata's `regress ... i.economycode, vce(cluster economycode)`.
- **Open item: age = 100.** This value looks like a special code. The 6 respondents in the common sample who have it are currently kept. Dropping them changes β_W in Stage 1, specification (2), by 0.02 percentage points.

STAGE 1

Reported scam contact

Question	Do more frequent payers report scam contact by phone or text more often?
Purpose	Exposure margin (H1). Also the comparison baseline for the placebo.
Sample	The whole common sample.
Outcome	con21 : 1 → 1, 2 → 0.

16,251
respondents

75
economies

4,770
with outcome = 1

75
economies with at least one
outcome = 1

Table 1.1 Main results

	(1) Country FE only	(2) Main: demographics	(3) Plus digital engagement
P_W weekly	31.00	30.55	30.53
P_M monthly	28.56	28.77	28.76
P_L less than monthly	24.59	25.19	25.24
β_W weekly – less than monthly	+6.41 [+4.06, +8.77] $p < 0.001$	+5.36 [+2.88, +7.83] $p < 0.001$	+5.29 [+2.80, +7.78] $p < 0.001$
β_M monthly – less than monthly	+3.97 [+1.60, +6.34] $p = 0.001$	+3.58 [+1.21, +5.96] $p = 0.004$	+3.51 [+1.11, +5.91] $p = 0.005$
$\beta_W - \beta_M$ weekly – monthly	+2.44 [+0.23, +4.65] $p = 0.031$	+1.77 [−0.60, +4.15] $p = 0.142$	+1.77 [−0.59, +4.14] $p = 0.140$

Percentage points. 95% confidence intervals in brackets. P_W , P_M and P_L are predicted probabilities: everyone's payment frequency is set to one level, all other variables keep their observed values, and the weighted mean is taken. Specification (2) is the main result.

Table 1.2 Raw weighted rates by payment frequency

Payment frequency	N	Outcome = 1	Weighted rate (%)
Weekly	7,358	2,138	29.16
Monthly	4,761	1,396	28.74
Less than monthly	4,132	1,236	27.82

No controls and no fixed effects. These rates pool all economies, so they also reflect differences between economies. Table 1.1 compares people within the same economy.

Table 1.3 All coefficients

	(1)	(2)	(3)
Weekly payer	+6.41*** (1.18)	+5.36*** (1.24)	+5.29*** (1.25)
Monthly payer	+3.97*** (1.19)	+3.58*** (1.19)	+3.51*** (1.20)
Age		+0.55*** (0.20)	+0.53*** (0.20)
Age squared / 100		−0.62** (0.24)	−0.58** (0.25)
Female		−2.69*** (1.01)	−2.60** (1.01)
Secondary education		−0.29 (1.54)	−0.91 (1.50)
Tertiary education		+2.18 (1.78)	+1.48 (1.75)
Income quintile 2		−2.00 (1.82)	−2.09 (1.82)
Income quintile 3		−2.02 (1.88)	−2.11 (1.88)
Income quintile 4		−2.15 (1.52)	−2.30 (1.50)
Income quintile 5		−2.35 (1.78)	−2.56 (1.79)
In the labor force		+4.78*** (1.12)	+4.65*** (1.12)
Rural		−1.16 (1.29)	−1.10 (1.30)
Uses the internet			+5.50*** (1.95)
Owns a smartphone			−1.74 (2.10)
Has an account			+5.12 (3.81)
Country fixed effects	Yes	Yes	Yes

Percentage points. Standard errors clustered by economy in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.10. Reference groups: less than monthly, primary education or less, income quintile 1. N = 16,251, 75 economies.

STAGE 2

Sending money, conditional on reported contact

Selected sample. Every result in this section is conditional on reported contact.

Question	Among respondents who were contacted, is the probability of sending money higher or lower for more frequent payers?
Purpose	Conditional vulnerability margin. H2a familiarity against H2b convenience, two-sided.
Sample	Common sample with <code>con21</code> = 1 and <code>con22</code> in {1, 2}.
Outcome	<code>con22</code> : 1 → 1, 2 → 0; 8 and 9 dropped.

4,768
respondents

75
economies

445
with outcome = 1

68
economies with at least one
outcome = 1

Table 2.1 Main results

	(1) Country FE only	(2) Main: demographics	(3) Plus digital engagement
P_W weekly	9.31	9.34	9.35
P_M monthly	10.03	10.03	10.08
P_L less than monthly	7.71	7.64	7.57
β_W weekly – less than monthly	+1.60 [–1.07, +4.27] p = 0.237	+1.71 [–1.03, +4.44] p = 0.218	+1.78 [–0.96, +4.52] p = 0.199
β_M monthly – less than monthly	+2.32 [–0.55, +5.19] p = 0.112	+2.39 [–0.51, +5.30] p = 0.105	+2.51 [–0.41, +5.43] p = 0.091
$\beta_W - \beta_M$ weekly – monthly	–0.72 [–3.34, +1.90] p = 0.585	–0.69 [–3.21, +1.84] p = 0.589	–0.73 [–3.25, +1.79] p = 0.566

Percentage points. 95% confidence intervals in brackets. P_W , P_M and P_L are predicted probabilities: everyone's payment frequency is set to one level, all other variables keep their observed values, and the weighted mean is taken. Specification (2) is the main result.

Table 2.2 Raw weighted rates by payment frequency

Payment frequency	N	Outcome = 1	Weighted rate (%)
Weekly	2,136	184	8.12
Monthly	1,396	145	10.29
Less than monthly	1,236	116	9.73

No controls and no fixed effects. These rates pool all economies, so they also reflect differences between economies. Table 2.1 compares people within the same economy.

Table 2.3 All coefficients

	(1)		(2)		(3)	
Weekly payer	+1.60	(1.34)	+1.71	(1.37)	+1.78	(1.37)
Monthly payer	+2.32	(1.44)	+2.39	(1.46)	+2.51*	(1.47)
Age			−0.13	(0.22)	−0.14	(0.22)
Age squared / 100			+0.17	(0.26)	+0.17	(0.26)
Female			−0.84	(0.96)	−0.82	(0.96)
Secondary education			−0.89	(1.75)	−0.76	(1.73)
Tertiary education			−2.65	(1.67)	−2.43	(1.64)
Income quintile 2			−5.11*	(2.73)	−5.17*	(2.74)
Income quintile 3			−6.47**	(2.51)	−6.42**	(2.51)
Income quintile 4			−5.43***	(2.00)	−5.39***	(2.02)
Income quintile 5			−4.86*	(2.53)	−4.72*	(2.57)
In the labor force			+3.57***	(1.26)	+3.48***	(1.26)
Rural			+0.22	(1.14)	+0.15	(1.14)
Uses the internet					+1.58	(2.81)
Owns a smartphone					−3.12	(2.39)
Has an account					+2.91	(4.49)
Country fixed effects	Yes		Yes		Yes	

Percentage points. Standard errors clustered by economy in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.10. Reference groups: less than monthly, primary education or less, income quintile 1. N = 4,768, 75 economies.

STAGE 3

Reported contact and sending money (joint event)

Question	Overall, is the probability of being contacted and sending money higher or lower for more frequent payers?
Purpose	Total risk (H3). Estimated directly, not as Stage 1 × Stage 2.
Sample	Common sample without the respondents who were contacted and answered con22 with 8 or 9.
Outcome	1 = contacted and sent money; 0 = not contacted, or contacted but did not send.

16,249
respondents

75
economies

445
with outcome = 1

68
economies with at least one
outcome = 1

Table 3.1 Main results

	(1) Country FE only	(2) Main: demographics	(3) Plus digital engagement
P_W weekly	2.92	2.91	2.92
P_M monthly	2.89	2.90	2.89
P_L less than monthly	1.75	1.75	1.75
β_W weekly – less than monthly	+1.17 [+0.24, +2.09] p = 0.014	+1.16 [+0.23, +2.10] p = 0.016	+1.17 [+0.23, +2.11] p = 0.015
β_M monthly – less than monthly	+1.14 [+0.14, +2.14] p = 0.026	+1.14 [+0.15, +2.14] p = 0.025	+1.14 [+0.13, +2.15] p = 0.028
$\beta_W - \beta_M$ weekly – monthly	+0.03 [−0.86, +0.91] p = 0.950	+0.02 [−0.89, +0.93] p = 0.967	+0.03 [−0.88, +0.94] p = 0.943

Percentage points. 95% confidence intervals in brackets. P_W , P_M and P_L are predicted probabilities: everyone's payment frequency is set to one level, all other variables keep their observed values, and the weighted mean is taken. Specification (2) is the main result.

Table 3.2 Raw weighted rates by payment frequency

Payment frequency	N	Outcome = 1	Weighted rate (%)
Weekly	7,356	184	2.37
Monthly	4,761	145	2.96
Less than monthly	4,132	116	2.71

No controls and no fixed effects. These rates pool all economies, so they also reflect differences between economies. Table 3.1 compares people within the same economy.

Table 3.3 All coefficients

	(1)	(2)	(3)
Weekly payer	+1.17** (0.46)	+1.16** (0.47)	+1.17** (0.47)
Monthly payer	+1.14** (0.50)	+1.14** (0.50)	+1.14** (0.51)
Age		−0.03 (0.06)	−0.03 (0.06)
Age squared / 100		+0.05 (0.08)	+0.05 (0.08)
Female		−0.43 (0.32)	−0.41 (0.32)
Secondary education		−0.62 (0.55)	−0.69 (0.57)
Tertiary education		−0.77 (0.56)	−0.85 (0.57)
Income quintile 2		−1.64* (0.89)	−1.64* (0.90)
Income quintile 3		−2.12*** (0.77)	−2.09*** (0.78)
Income quintile 4		−1.91*** (0.70)	−1.89** (0.72)
Income quintile 5		−1.68* (0.90)	−1.64* (0.92)
In the labor force		+1.38*** (0.37)	+1.32*** (0.37)
Rural		+0.11 (0.33)	+0.10 (0.33)
Uses the internet			+1.72 (1.04)
Owns a smartphone			−1.56 (1.07)
Has an account			+1.99 (1.38)
Country fixed effects	Yes	Yes	Yes

Percentage points. Standard errors clustered by economy in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.10. Reference groups: less than monthly, primary education or less, income quintile 1. N = 16,249, 75 economies.